Kept

A guarded next best action for promises customers can keep.

Team Sharpe Shooters • CIBC Collections Hackathon • 4 Oct 2026

1M
customers,
one golden ID each

96,000
guarded actions
a night

62.5%
vs 51.5% of promised
money paid when dated
to payday



Promises set up to break

48.8%

of cases entering at 1-30 days past due roll by day 30

and rolling is the only road to charge-off

2,352

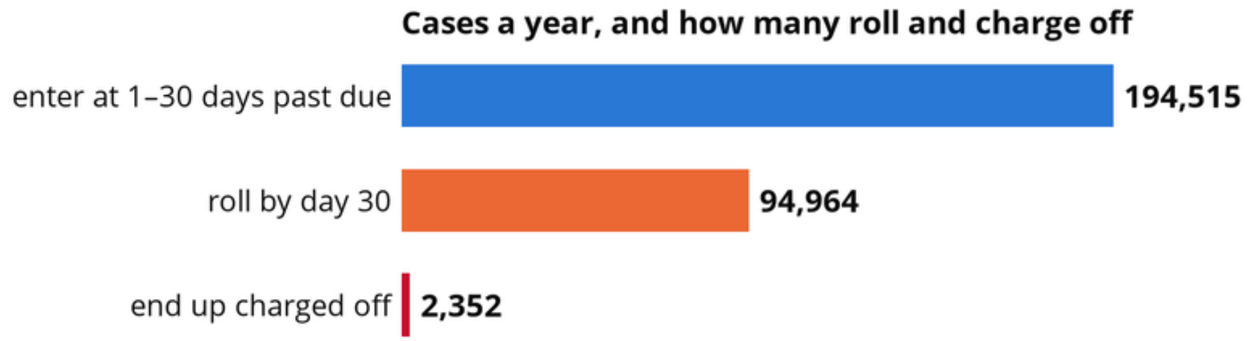
charge-offs a year

One person, five IDs across 31 tables

The facts live in free-text notes

Promises fall due before payday

Hardship found too late



Rolling is the only road to a charge-off: 2.5% of rolled cases charge off (average $12.5K), 0% of the rest.

When you ask matters as much as who you ask

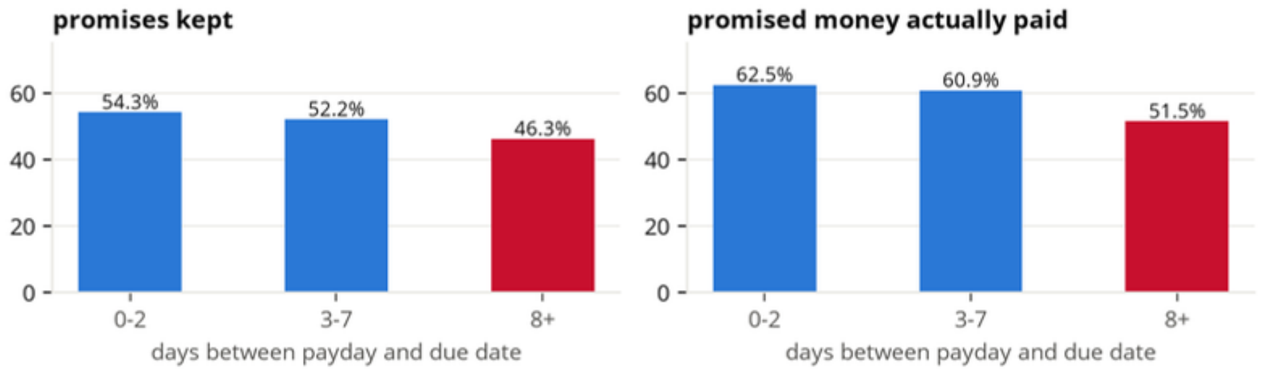
62.5%

of promised money paid when the promise is dated 0-2 days after payday

51.5%

of promised money paid when the promise is dated 8+ days after payday

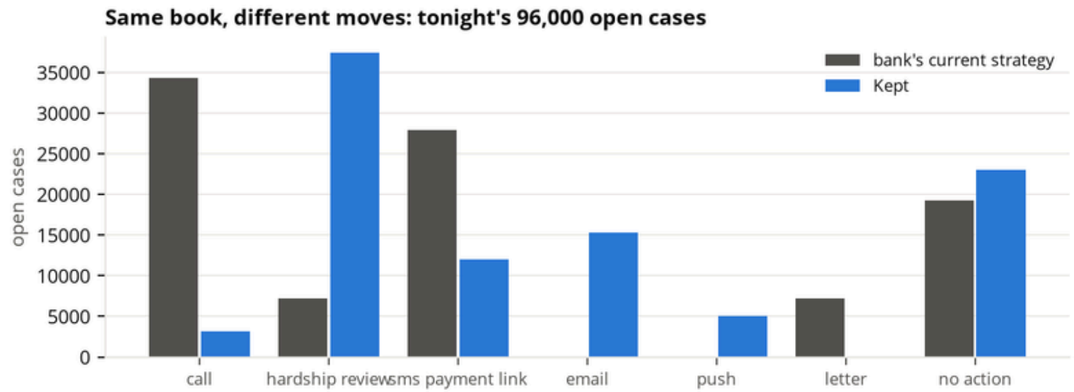
Based on 32,000 resolved promises with visible salary deposits



So Kept dates every promise 1–2 days after the customer's payday, sized to their spare cash.

One card per case

Rules first, then an uplift model, then a promise sized to payday. Every card shows three plain-English reasons.



1 → **2** → **3** → **4** → **5** → **6** → **7**

Hard stops
Block ineligible cases

Permissions
Check consent and access

Hardship rules
Apply policy rules (no AI)

Uplift model
Estimate impact of actions

Payday sizing
Choose right timing and size

Rank
Order actions by value

Explain
Show reason for action

Call targeting (Qini)

4x

vs the bank's current strategy

0.0073 vs 0.0017

ACTION TIERS

GREEN	AMBER	RED
Self-serve	Needs an agent	Needs a specialist plus a team lead

A PERSON RELEASES EVERY ACTION

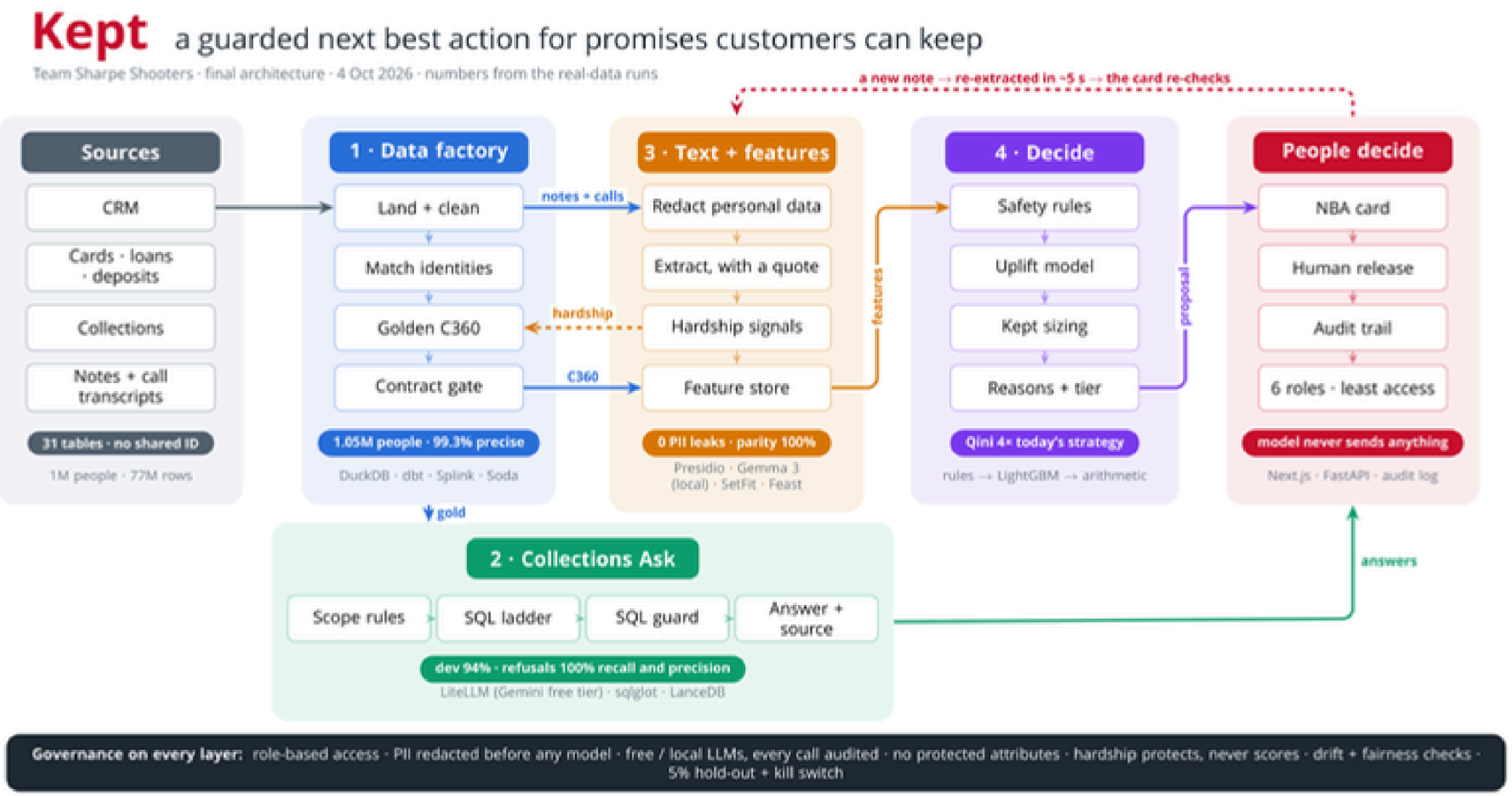
TODAY'S PROCESS

5,328
contacts outside calling hours

290
7-day contact-cap breaches (121 cases)

Kept: 0 of both, by construction

Four layers, governance through all of them



Our own TikZ diagram. Governance runs through every layer.

Data you can trust

68.2M

rows across 31 tables

3.84M → 1.05M

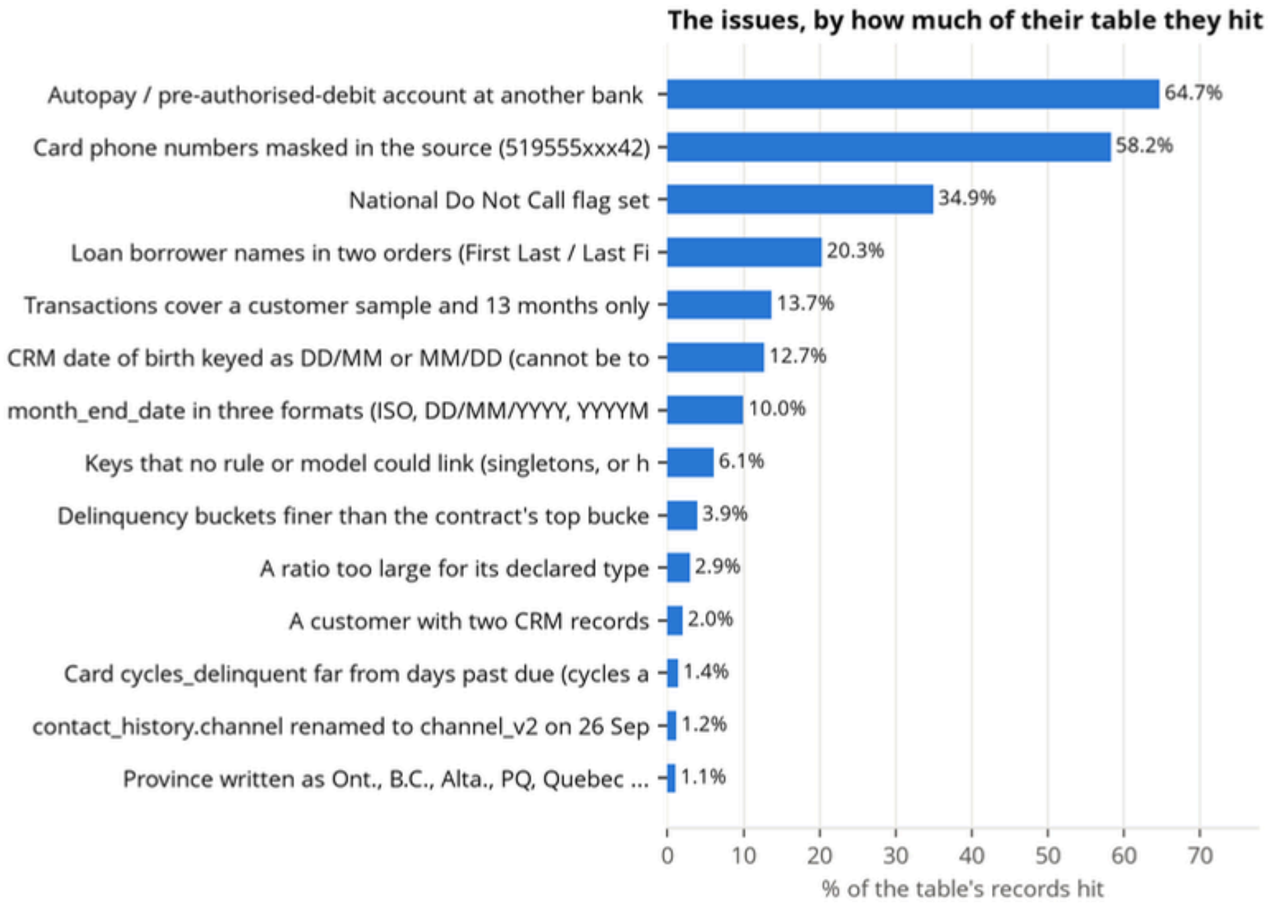
source keys to golden IDs at 99.3% precision

19

data-quality issues counted and fixed

PASS

contract gate on all 8 rules



Every issue counted and handled with a tested rule; nothing guessed. The contract gate, built from DC-COLL-001, blocks a bad load.

Ask in plain English, get answers with their working

20 of 21

dev benchmark questions right (17 of 18 in scope, 3 of 3 refusals)

100%

refusal recall

100%

refusal precision

Every answer shows its SQL or a cited source you can check.

Protected-attribute questions refused before any LLM sees them.

How a question gets answered

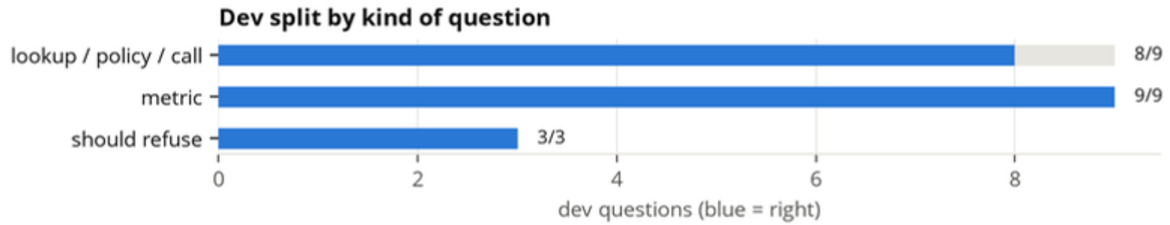
1 Rules first: refuse protected, bulk-PII and write questions

2 Certified metrics and SQL templates; dates worked out in code

3 Free SQL last: three drafts, majority vote

4 Guard: SELECT only, on 17 approved tables

5 Every number in the answer must appear in the result rows



Free LLMs only (Gemini free tier for the Ask, Gemma 3 run locally for notes); every call is audited.

Fair by construction

0 of 96,000

decisions change when every protected attribute is scrambled

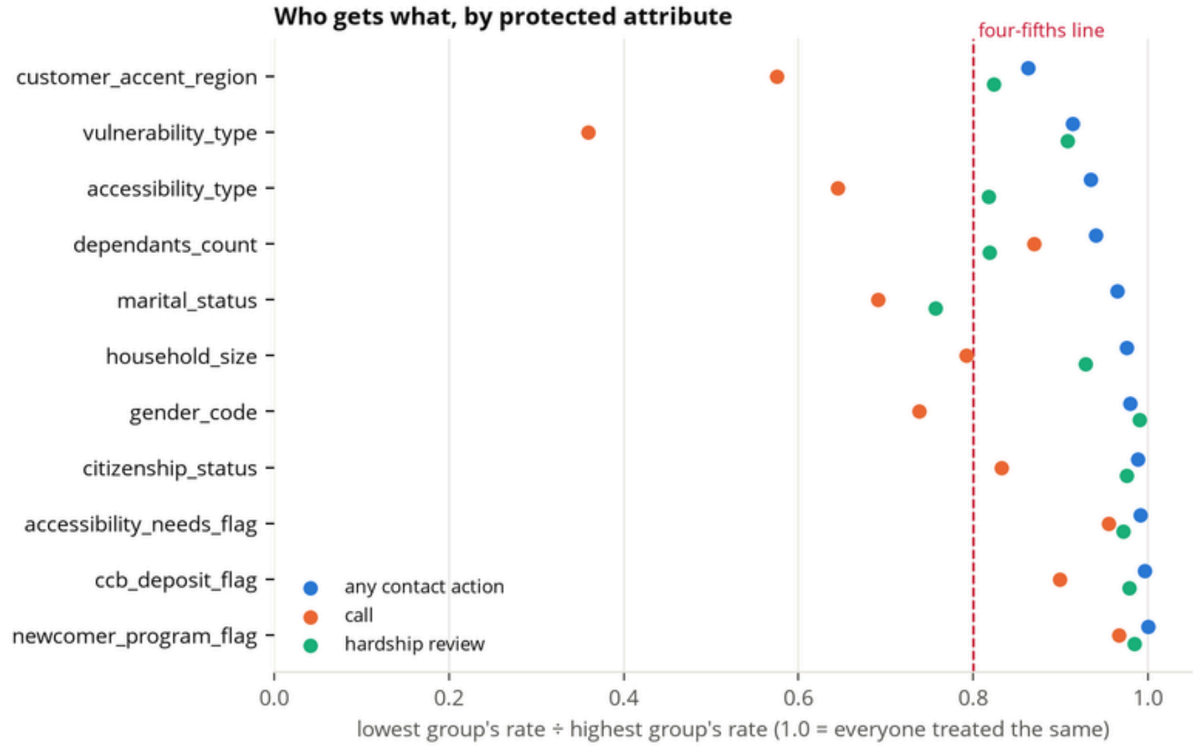
0.86–1.00

any-contact ratio across every protected attribute (four-fifths line 0.80)

scrambling proxies too moves

299 cards

all toward more protection



Blue = any contact action, all at 0.86 or above. The flags sit on rare calls and on protective hardship reviews.

$2.2M a year, paid back in about 2 months

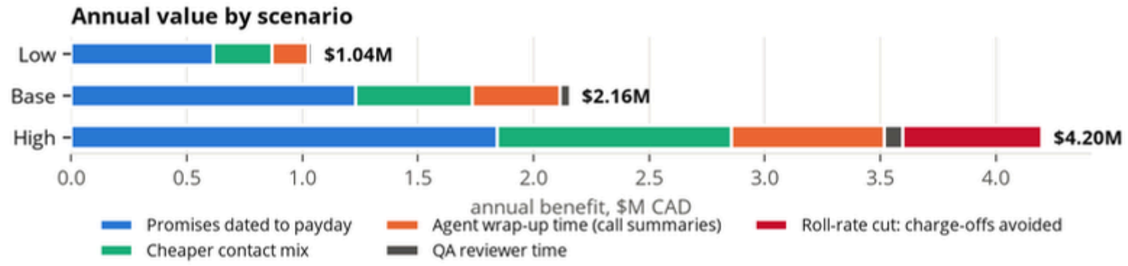
$2.2M a year ($1.0M–4.2M)

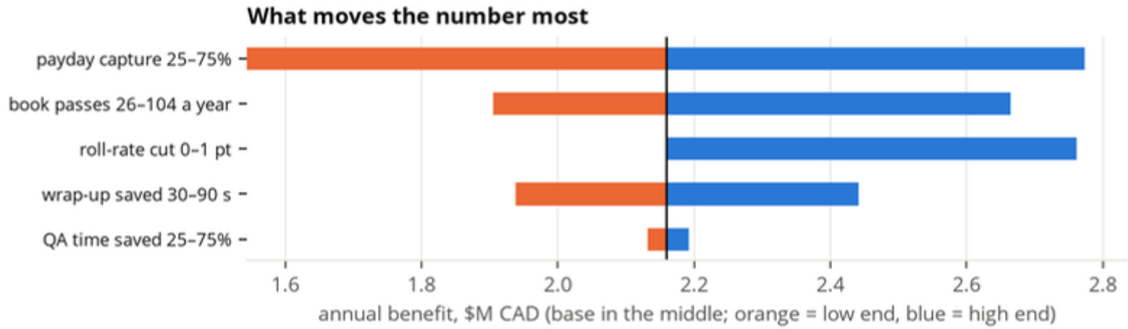
3.8x year-one ROI

~2 months payback

$300K build · $150K a year to run

Launch with a 20% hold-out: it detects a 2-point lift in about 2 months (5% would take 6+ months), then shrink to the permanent 5%. A kill switch reverts every card.





Every point of 30-day roll rate is worth about $0.6M a year in charge-offs avoided, not counted in the base case.

Built together. Promises customers can keep.

4 tracks · 68 PRs · 335 commits · 5 integration gates

TEAM

A Data and identity
Satvik Rao

B Ask and fairness
Garimella Sai Abhishek

C Text, hardship and GPU
Akshat Srivastava

D NBA engine and web app
Shreyas Deb

NEXT STEPS

Retrain with payroll features

Plan-aware rule: 19% fewer Red reviews (7,355 of 38,115 are already on a plan)

Read the hold-out at day 30

Promises customers can keep.